Himanshu

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Education

VIT Bhopal University — Bhopal, Madhya Pradesh, India
B.Tech. in Computer Science (AI & ML) | CGPA: 9.01/10 — *Sept. 2023 – May. 2027*

Experience

LLM Engineering Intern — Mar. 2025 – Sept. 2025
amasQIS.ai — *Remote*

- Engineered an **LLM-powered product recommendation engine** by fine-tuning **Mistral-7B** using parameter-efficient **LoRA/PEFT** adaptation, improving recommendation relevance by **15%**.
- Orchestrated a high-throughput **data preprocessing** and **prompt engineering** workflow over **50,000+ records**, applying feature engineering and inference optimization to reduce data noise by **40%** and improve LLM workflow efficiency.
- Operationalized production-ready **AI model serving** pipelines with **MLOps** best practices for model versioning, performance benchmarking, and latency-aware model optimization — ensuring scalable, reliable **AI system deployment** in a cloud-hosted environment.
- **Technologies:** Python, Mistral-7B, LoRA/PEFT, Prompt Engineering, MLOps, Model Serving

Projects

Semantic Search & Vector Retrieval Engine | *Sentence-Transformers, FAISS, GMM, FastAPI* — **GitHub**

- Architected a production-grade **Semantic Search & Vector Retrieval Engine** over **19,125 documents** using **FAISS** vector databases and **all-MiniLM-L6-v2** embedding models, enabling scalable **information retrieval**.
- Implemented an **unsupervised clustering** architecture using **BIC-optimized GMM** (k=30) with **PCA dimensionality reduction** (384D $\rightarrow$ 50D), achieving **88.8%** high-confidence cluster assignments across 20 Newsgroups and facilitating structured **semantic routing** over noisy real-world text data.
- Designed a two-layer **semantic caching** and **intelligent retrieval routing** mechanism, achieving a **1,297× latency speedup** (28ms $\rightarrow$ 22μs) through cluster-aware **inference optimization**.

AI Financial Decision Support Agent | *FinBERT, LangChain, FAISS, Gemini, RAG* — **GitHub**

- Built an enterprise-ready **AI Financial Decision Support Agent** integrating **FinBERT** sentiment analysis and **PyTorch** forecasting models for real-time financial analytics, achieving $\mathbf{R^2 = 0.92}$ and **RMSE** $\approx$ **2.5%** on held-out market data.
- Integrated a **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain**, **FAISS**, and **all-MiniLM-L6-v2** embeddings with **Google Gemini**, enabling natural language question answering and intelligent document retrieval over enterprise financial corpora.
- Delivered an end-to-end **AI workflow** spanning data ingestion, model training, and inference orchestration through an interactive **Streamlit** dashboard, enabling real-time forecasting and **LLM-powered** financial insight generation.

Technical Skills

Languages: Python, C++, SQL
Generative AI: LLMs, LLM Fine-Tuning, RAG, Prompt Engineering, AI Agents, LoRA/PEFT, Mistral-7B
Machine Learning: Deep Learning, Natural Language Processing (NLP), Information Retrieval, Embedding Models, PyTorch, FinBERT, GMM Clustering, PCA Dimensionality Reduction
Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, FastAPI, Sentence-Transformers, Hugging Face
AI Infrastructure: FAISS, Vector Databases, Knowledge Graphs, MLOps, Model Serving
Developer Tools: Git, GitHub, Jupyter Notebook, Streamlit
Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming, Computer Networks

Achievements & Certifications

- Secured a position among the **Top 6 Teams** at **HackSethu**, a national-level hackathon, by developing an AI-driven solution under real-world constraints.
- Active **Core Member** at **Google Developer Group (GDG)**, contributing to community-driven technical initiatives and AI/ML knowledge sharing.
- Completed the **GEN AI Using IBM Watsonx** and **Microsoft Certified: Azure Data Fundamentals** certifications.
- Solved **300+ Data Structures & Algorithms** problems on LeetCode, demonstrating strong algorithmic and problem-solving skills.